

Name: _____

Name the criterion (SSS, SAS, ASA, AAS, or HL) that the given information provides.

- 1. Three pairs of congruent sides.
- 2. Two pairs of congruent angles and the pair of congruent sides between them.
- 3. Two pairs of congruent sides and the pair of congruent angles between them.
- 4. Two right triangles with congruent hypotenuses and one pair of congruent legs.
- 5. Two pairs of congruent angles and a pair of congruent sides that is NOT between them.

Solve.

- 6. $\triangle ABC \cong \triangle DEF$, $AB = 3x + 2$, and $DE = 20$. Find x .
- 7. $\triangle MNO \cong \triangle PQR$, $MN = 7y - 5$, and $PQ = 30$. Find y and MN .

Answer each question.

- 8. Two triangles have two pairs of congruent sides and a pair of congruent angles that is NOT between those sides. Is that enough to prove them congruent? Explain.
- 9. How is HL different from SSA, given that HL also uses two sides and a non-included angle?

Complete the two-column proof.

10. **Given:** $\overline{QP}$ bisects $\angle RQS$; $\overline{QR} \cong \overline{QS}$ **Prove:** $\triangle RQP \cong \triangle SQP$

Statements	Reasons
1. $\overline{QP}$ bisects $\angle RQS$	1. Given
2.	2.
3.	3.
4.	4.
5.	5.

